

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID: SIH26166

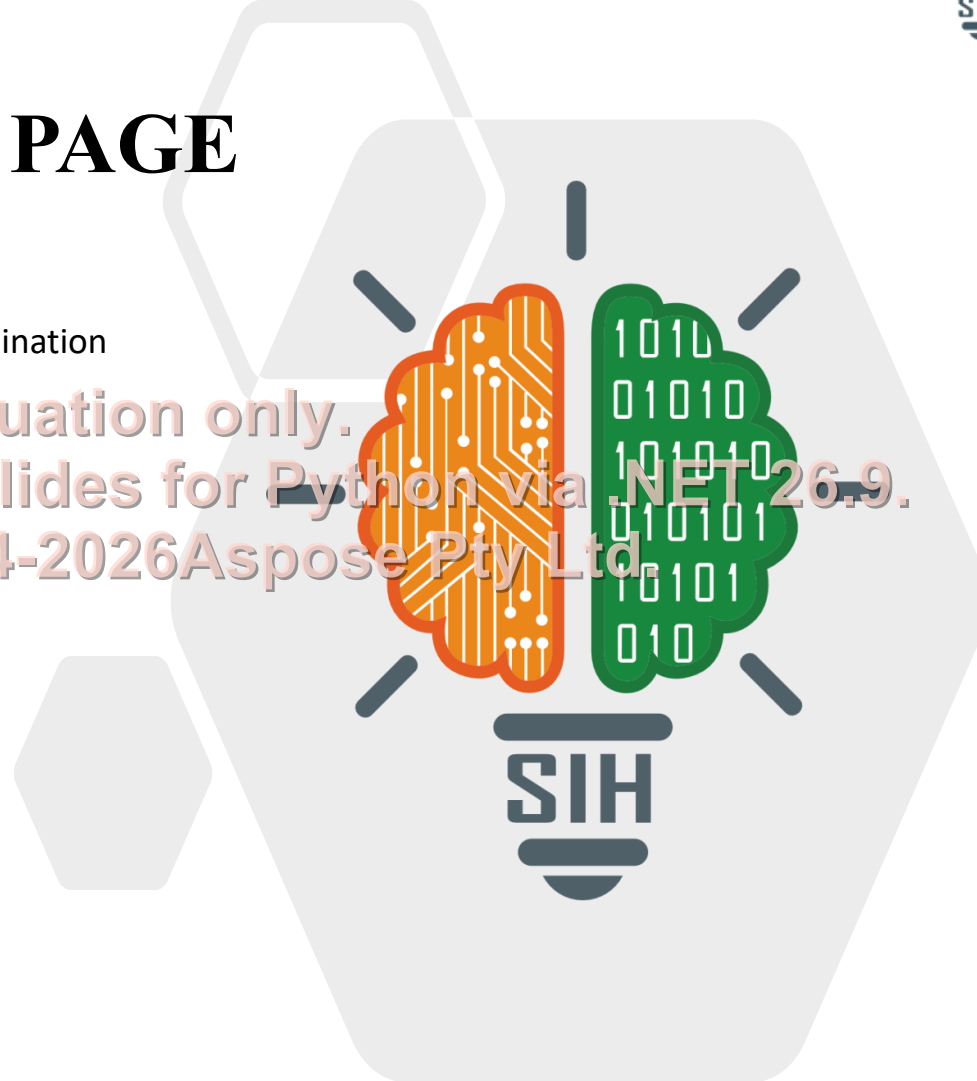
Problem Statement Title: Lunar Image Registration Under Extreme Illumination

Variations

Theme: Space Technology | PS Category: Software

Team ID: 185903 | Team Name: Maximus2

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LunaProof: Multi-Modal Illumination Invariant Lunar Image Registration



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❖ Proposed Solution & Mission-Grade Innovation

- Physics-Based Lunar Rendering: Lommel-Seeliger scattering + ray-marched cast shadows for true illumination modeling.
- Illumination-Invariant Phase Congruency: Kovesi log-Gabor filters extract edge structures invariant to sun-angle flips.
- Multi-Matcher Framework: Benchmarking SIFT, PC-SIFT, LoFTR (zero-shot), and custom PatchNet descriptors.
- Ground-Truth Checkpoint Metrology: Measures sub-pixel error against exact ground truth at held-out checkpoints (not fitted matches).
- Autonomous Refusal Gate: Observes match count, inlier ratio, and spatial coverage to refuse un-reliable registrations.

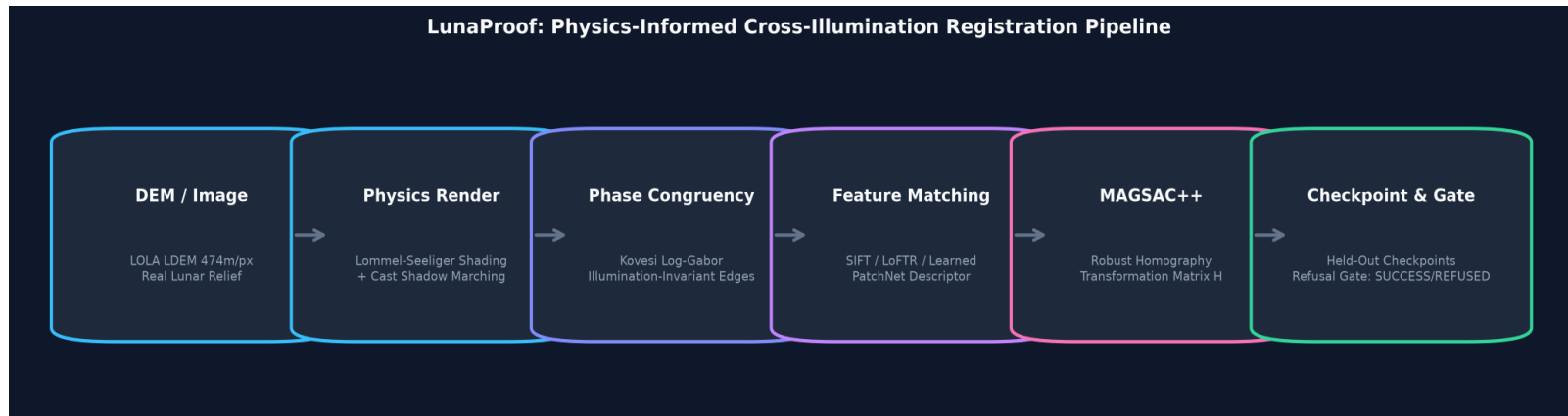
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TECHNICAL APPROACH



- **Technical Architecture & Pipeline**

- Tech Stack: Python, PyTorch, OpenCV, NumPy, SciPy, Kornia, Google Colab, GitHub.
- Dataset: NASA LRO LOLA LDEM_64 elevation tiles (64 PPD, ~474 m/px) split strictly by lunar region.
- Pipeline: DEM Tile -> Lommel-Seeliger + Cast Shadows -> Log-Gabor Phase Map -> Feature Matcher -> MAGSAC++ -> Refusal Gate.



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FEASIBILITY AND VIABILITY



- **Rigorous Held-Out Benchmark & Competitive Contrast**

- Full Sun-Angle Sweep (0° – 180°): Unlike single-orbit demos (0.76° delta), we benchmark across 15 non-overlapping regions (112 test pairs).
- SIFT Baseline: Measured 28.6% overall success rate (0.0% at 180° sun gap).
- Phase-Congruency SIFT: Measured 88.4% overall success rate (100.0% at 180° sun gap).
- LoFTR (Zero-Shot): Measured 73.2% overall success rate.
- Learned PatchNet Descriptor: Measured 98.2% overall success rate in our held-out test.
- Mission-Grade Refusal Gate: Accepted pairs were correct 93.6% of the time, automatically rejecting degraded inputs.
- Honest Failure Analysis: Under sun gaps $\geq 60^{\circ}$, SIFT drops to 0.0%. Median error showcase: 0.685 px.

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IMPACT AND BENEFITS



- **ISRO Mission Impact & Multi-Scale Roadmap**

- Multi-Modal Co-Registration: Enables robust registration between Chandrayaan-2 OHRC (0.25 m), TMC-2 (5m), and IIRS (~80m).
- Safety-Critical Refusal: Prevents corrupted GIS mapping by automatically rejecting ill-conditioned feature matches.
- Scale Cascade Roadmap: IIRS 80m -> TMC-2 5m -> LRO NAC 1m -> OHRC 0.25m (hop ratios 16x, 5x, 4x).

Multi-Modal Resolution Cascade Architecture (Roadmap)



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- Kovesi, P. (1999). Image Features from Phase Congruency. Videre: J. Comput. Vis. Res., 1(3), 1-26.
- Sun, J., et al. (2021). LoFTR: Detector-Free Local Feature Matching with Transformers. CVPR 2021.
- Baranjague et al. (2020). USAC: A Universal Framework for Random Sample Consensus. IEEE TPAMI.
 - NASA LRO LOLA LDEM_64 Global Lunar Digital Elevation Model (NASA PDS Geosciences Node).
 - ISRO Chandrayaan-2 OHRC, TMC-2 & IIRS Payload Specifications.
 - GitHub Repository & Executable Notebook: github.com/pintocyrus/LunaProof



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- **Primary Research Citations & Repository**

- Kovesi, P. (1999). Image Features from Phase Congruency. Videre: J. Comput. Vis. Res., 1(3), 1-26.
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